SAHIL JASUJA

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SUMMARY

Software Engineer with 2+ years building large-scale C++ systems, parser infrastructure, distributed checkpoint-restart systems, and AI-assisted developer tooling (MCP) at IBM Systems. Experienced in performance optimization, ANTLR-based parser design, Linux systems programming, and DevOps/CI-CD automation across a 250K+ LOC production codebase. M.Tech, Computer Science, IIT Madras; GATE CSE 2022 AIR 94 (top 0.1% nationwide).

TECHNICAL SKILLS

- **Languages:** C++, C, Go, Python, JavaScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Design, Design Patterns, Low-Level Design, System Design, Concurrency, Distributed Systems
- **Modern C++:** STL, Templates, Multithreading, Memory Management, Performance Optimization, GDB
- **Compiler & Parser:** ANTLR, Grammar Migration, Visitor Pattern, Parser Development
- **AI & Developer Tooling:** MCP (Model Context Protocol), Agentic AI, Prompt Engineering, AI-Assisted Developer Tooling
- **DevOps & Backend:** CI/CD, Release Engineering, Build & Regression Pipelines, Git, Docker, Linux, Google Test, Hyperledger Fabric, React Native

EXPERIENCE

IBM Systems — Software Engineer Jul 2024 – Present
Bengaluru, India

- Developed performance-critical C++ components for a **250K+ LOC C++** large-scale, simulation-infrastructure codebase supporting **300+ internal engineers**, contributing through quarterly production releases.
- Migrated a production parser from **ANTLR v1 to ANTLR 4.9** using the visitor pattern, reducing grammar complexity from **4,000+ lines to ~500 lines** (a **~87% reduction**) while preserving full backward compatibility across an internal DSL.
- Extended an internal **constraint-specification language (CSP/DSL)** used for simulation control by implementing new language constructs, parser enhancements, and semantic correctness fixes for **300+ engineers**.
- Designed and implemented a Distributed MultiThreaded Checkpoint-Restart (**DMTCP**) system, reducing recomputation overhead by **~25%** and saving significant compute time across verification workflows.
- Built a modular configuration engine with macro expansion, validation, and dependency resolution, eliminating manual config errors across **300+ users**; improved resolution performance by **~20%** via profiling-driven optimization.
- Built an **MCP** (Model Context Protocol)-based **agentic AI developer assistant** capable of translating natural-language requirements into CSP configuration code using tool orchestration and reusable skills.
- Improved **release engineering** workflows by strengthening **CI/CD** validation, automated regression pipelines, and release gating for quarterly production releases supporting **300+ engineers**.
- Designed an **AI-assisted migration pipeline** — gated stages with automated failure classification — for converting legacy **CppUnit** suites to **GoogleTest**, integrating automated validation into CI to improve regression coverage by **~40%**.
- Led the **cross-platform port** of the DMTCP checkpoint-restart system to a new CPU architecture (**PPC64**), debugging and fixing low-level **assembly/ABI** mismatches and a coordinator-barrier deadlock to unlock cross-platform recomputation savings.

Samsung Research Institute — Technical Intern May 2023 – Jul 2023
Bengaluru, India

- Implemented and evaluated Generative Adversarial Network (**GAN**) models across **1,000+ image samples**, analyzing convergence and stability across multiple training configurations.
- Improved the data preprocessing pipeline, reducing manual dataset preparation effort by **~30%**.

PROJECTS

Distributed Marketplace for Used Goods — M.Tech Thesis, IIT Madras Jul 2023 – May 2024

- Designed and implemented a permissioned blockchain-based marketplace using **Hyperledger Fabric**, architected to support **10,000+ users** with secure, consistent distributed state.
- Developed backend services and chaincode in **Go** to manage transaction validation and distributed ledger state.
- Integrated a **React Native** mobile application as the client interface for the marketplace.

EDUCATION

Indian Institute of Technology Madras — M.Tech, Computer Science Jul 2022 – Jun 2024
CGPA: 8.38/10 | Chennai, India

MediCaps University — B.Tech, Computer Science Jul 2018 – May 2022
CGPA: 8.56/10 | Indore, India

ACHIEVEMENTS

- GATE CSE 2022 — **All India Rank 94** (Top 0.1% nationwide)